INCH-POUND
MS35764G
17 June 2021
SUPERSEDING
MS35764F
29 May 2013

DETAILED SPECIFICATION SHEET

BOLT, SELF-LOCKING, HEXAGON HEAD, STEEL, GRADE 8, UNC-2A AND UNF-2F

Reactivated after 17 June 2021 and may be used for new and existing designs and acquisitions.

This specification sheet is approved for all Departments and Agencies of the Department of Defense.

The requirements for acquiring the described herein shall consist of this specification sheet and procurement specification MIL-DTL-45912.

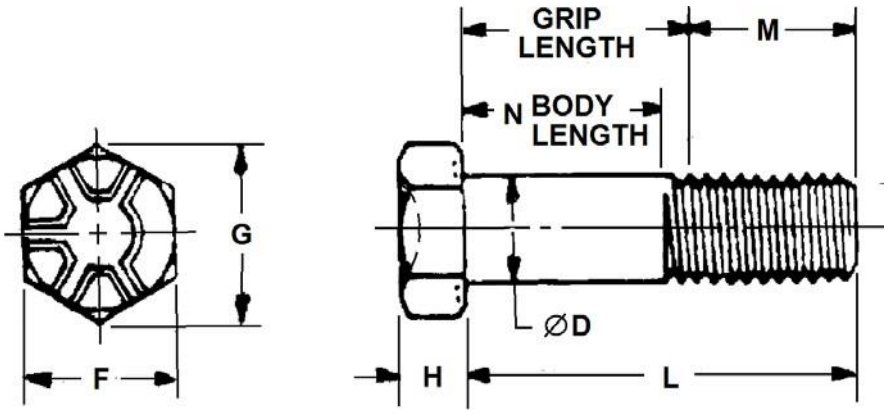


FIGURE 1. Bolt, self-locking.

Comments, suggestions, or questions on this document should be addressed to U.S. Army CCDC, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-IES, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.ccdc-devcom.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5306

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TABLE I. Part numbers and dimensions.

NOMINAL SIZE		0.25				0.3125				0.375				0.5			
THREADS PER INCH		20	28			18	24			16	24			13	20		
BODY MAX ØD DIAMETER MIN		0.25 0.245				0.3125 0.3065				0.375 0.369				0.500 0.493			
ACROSS F FLATS	MAX MIN	0.500 0.489				0.562 0.551				0.625 0.612				0.812 0.798			
ACROSS G CORNERS	MAX MIN	0.577 0.557				0.650 0.628				0.722 0.698				0.936 0.910			
HEAT H HEIGHT	MAX MIN	0.135 0.125				0.198 0.188				0.260 0.250				0.327 0.312			
THREAD M LENGTH	REF	0.750				0.875				1.000				1.250			
UNTHREADED * N BODY LGT	MAX	0.125	0.089			0.139	0.104			0.156	0.104			0.192	0.125		
L LENGTH	FINISH /1	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX
0.75	CAD	1201	1223			1245	1267			1289	1311			1395	1425		
	PHOS	1202	1224			1246	1268			1290	1312			1396	1426		
	ZNNI-K16	1001	1101			1007	1107			1015	1115			1025	1125		
	ZNNI-K20	2001	2101			2007	2107			2015	2115			2025	2125		
0.875	CAD	1203	1225		0.125	1247	1269			1291	1313			1397	1427		
	PHOS	1204	1226			1248	1270			1292	1314			1398	1428		
	ZNNI-K16	1002	1102			1008	1108			1016	1116			1026	1126		
	ZNNI-K20	2002	2102			2008	2108			2016	2116			2026	2126		
1.000	CAD	1205	1227		0.250	1249	1271			1293	1315			1399	1429		
	PHOS	1206	1228			1250	1272			1294	1316			1400	1430		
	ZNNI-K16	1003	1103			1009	1109			1017	1117			1027	1127		
	ZNNI-K20	2003	2103			2009	2109			2017	2117			2027	2127		
1.250	CAD	1207	1229	0.100	0.500	1251	1273		0.375	1295	1317		0.250	1401	1431		
	PHOS	1208	1230			1252	1274			1296	1318			1402	1432		
	ZNNI-K16	1004	1104			1010	1110			1018	1118			1028	1128		
	ZNNI-K20	2004	2104			2010	2110			2018	2118			2028	2128		
1.500	CAD	1209	1231	0.350	0.750	1253	1275	0.208	0.625	1297	1319	0.062	0.500	1403	1433		0.250
	PHOS	1210	1232			1254	1276			1298	1320			1404	1434		
	ZNNI-K16	1005	1105			1011	1111			1019	1119			1029	1129		
	ZNNI-K20	2005	2105			2011	2111			2019	2119			2029	2129		
1.750	CAD	1211	1233	0.600	1.000	1255	1277	0.458	0.875	1299	1321	0.312	0.750	1405	1435	0.019	0.500
	PHOS	1212	1234			1256	1278			1300	1322			1406	1436		
	ZNNI-K16	1006	1106			1012	1112			1020	1120			1030	1130		
	ZNNI-K20	2006	2106			2012	2112			2020	2120			2030	2130		
2.000	CAD					1257	1279	0.708	1.125	1301	1323	0.562	1.000	1407	1437	0.269	0.750
	PHOS					1258	1280			1302	1324			1408	1438		
	ZNNI-K16					1013	1113			1021	1121			1031	1837		
	ZNNI-K20					2013	2113			2021	2121			2031	2838		
2.250	CAD					1259	1281	0.958	1.375	1303	1325	0.812	1.250	1409	1439	0.519	1.000
	PHOS					1260	1282			1304	1326			1410	1440		
	ZNNI-K16					1014	1114			1022	1122			1032	1132		
	ZNNI-K20					2014	2114			2022	2122			2032	2132		
2.500	CAD									1305	1327	1.062	1.500	1411	1441	0.769	1.250
	PHOS									1306	1328			1412	1442		
	ZNNI-K16									1023	1123			1033	1133		
	ZNNI-K20									2023	2123			2033	2133		
2.750	CAD									1307	1329	1.312	1.750	1413	1443	1.019	1.500
	PHOS									1308	1330			1414	1444		
	ZNNI-K16									1024	1124			1034	1134		
	ZNNI-K20									2024	2124			2034	2134		

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TABLE I. Part numbers and dimensions – Continued.

NOMINAL SIZE		0.25				0.3125				0.375				0.5			
THREADS PER INCH		20	28			18	24			16	24			13	20		
BODY MAX ØD DIAMETER MIN		0.25 0.245				0.3125 0.3065				0.375 0.369				0.500 0.493			
ACROSS F FLATS	MAX MIN	0.500 0.489				0.562 0.551				0.625 0.612				0.812 0.798			
ACROSS G CORNERS	MAX MIN	0.577 0.557				0.650 0.628				0.722 0.698				0.936 0.910			
HEAT H HEIGHT	MAX MIN	0.135 0.125				0.198 0.188				0.260 0.250				0.327 0.312			
THREAD M LENGTH	REF	0.750				0.875				1.000				1.250			
UNTHREADED /2 N BODY LGT	MAX	0.125	0.089			0.139	0.104			0.156	0.104			0.192	0.125		
L LENGTH	FINISH /1	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX
3.000	CAD PHOS ZNNI-K16 ZNNI-K20													1415 1416 1035 2035	1445 1446 1135 2135	1.269	1.750
3.250	CAD PHOS ZNNI-K16 ZNNI-K20													1417 1418 1036 2036	1447 1448 1136 2136	1.519	2.000
3.500	CAD PHOS ZNNI-K16 ZNNI-K20													1419 1420 1037 2037	1449 1450 1137 2137	1.769	2.250
3.750	CAD PHOS ZNNI-K16 ZNNI-K20													1421 1422 1038 2038	1451 1452 1138 2138	2.019	2.500

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TABLE I. Part numbers and dimensions – Continued.

NOMINAL SIZE		0.625				0.750				1.000				1.250			
THREADS PER INCH		20	28			18	24			16	24			13	20		
BODY MAX ØD DIAMETER MIN		0.25 0.245				0.3125 0.3065				0.375 0.369				0.500 0.493			
ACROSS F FLATS	MAX MIN	0.500 0.489				0.562 0.551				0.625 0.612				0.812 0.798			
ACROSS G CORNERS	MAX MIN	0.577 0.557				0.650 0.628				0.722 0.698				0.936 0.910			
HEAT H HEIGHT	MAX MIN	0.135 0.125				0.198 0.188				0.260 0.250				0.327 0.312			
THREAD M LENGTH	REF	0.750				0.875				1.000				1.250			
UNTHREADED * N BODY LGT	MAX	0.125	0.089			0.139	0.104			0.156	0.104			0.192	0.125		
L LENGTH	FINISH /1	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX
1.000	CAD	1515	1543														
	PHOS	1516	1544														
	ZNNI-K16	1039	1139														
	ZNNI-K20	2039	2139														
1.250	CAD	1517	1545			1571	1599										
	PHOS	1518	1546			1572	1600										
	ZNNI-K16	1040	1140			1053	1153										
	ZNNI-K20	2040	2140			2053	2153										
1.500	CAD	1519	1547			1573	1601										
	PHOS	1520	1548			1574	1602										
	ZNNI-K16	1041	1141			1054	1154										
	ZNNI-K20	2041	2141			2054	2154										
1.750	CAD	1521	1549		0.250	1575	1603			1679	1705						
	PHOS	1522	1550			1576	1604			1680	1706						
	ZNNI-K16	1042	1142			1055	1155			1066	1166						
	ZNNI-K20	2042	2142			2055	2155			2066	2166						
2.000	CAD	1523	1551		0.500	1577	1605		0.250	1681	1707			1729	1753		
	PHOS	1524	1552			1578	1606			1682	1708			1730	1754		
	ZNNI-K16	1043	1143			1056	1156			1067	1167			1078	1178		
	ZNNI-K20	2043	2143			2056	2156			2067	2167			2078	2178		
2.250	CAD	1525	1553		0.750	1579	1607		0.500	1683	1709			1731	1755		
	PHOS	1526	1554			1580	1608			1684	1710			1732	1756		
	ZNNI-K16	1044	1144			1057	1157			1068	1168			1079	1179		
	ZNNI-K20	2044	2144			2057	2157			2068	2168			2079	2179		
2.500	CAD	1527	1555	0.227	1.000	1581	1609		0.750	1685	1711			1733	1757		
	PHOS	1528	1556			1582	1610			1686	1712			1734	1758		
	ZNNI-K16	1045	1145			1058	1158			1069	1169			1080	1180		
	ZNNI-K20	2045	2145			2058	2158			2069	2169			2080	2180		
2.750	CAD	1529	1557	0.477	1.250	1583	1611	0.200	1.000	1687	1713		0.500	1735	1759		
	PHOS	1530	1558			1584	1612			1688	1714			1736	1760		
	ZNNI-K16	1046	1146			1059	1159			1070	1170			1081	1181		
	ZNNI-K20	2046	2146			2059	2159			2070	2170			2081	2181		
3.000	CAD	1531	1559	0.727	1.500	1585	1613	0.450	1.250	1689	1715		0.750	1737	1761		
	PHOS	1532	1560			1586	1614			1690	1716			1738	1762		
	ZNNI-K16	1047	1147			1060	1160			1071	1171			1082	1182		
	ZNNI-K20	2047	2147			2060	2160			2071	2171			2082	2182		
3.250	CAD	1533	1561	0.977	1.750	1587	1615	0.700	1.500	1691	1717	0.125	1.000	1739	1763		0.500
	PHOS	1534	1562			1588	1616			1692	1718			1740	1764		
	ZNNI-K16	1048	1148			1161	1161			1072	1172			1083	1183		
	ZNNI-K20	2048	2148			2161	2161			2072	2172			2083	2183		
3.500	CAD	1535	1563	1.227	2.000	1589	1617	0.950	1.750	1693	1719	0.375	1.250	1741	1765		0.750
	PHOS	1536	1564			1590	1618			1694	1720			1742	1766		
	ZNNI-K16	1049	1149			1062	1162			1073	1173			1084	1184		
	ZNNI-K20	2049	2149			2062	2162			2073	2173			2084	2184		

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TABLE I. Part numbers and dimensions – Continued.

NOMINAL SIZE		0.625				0.750				1.000				1.250			
THREADS PER INCH		20	28			18	24			16	24			13	20		
BODY ØD DIAMETER	MAX	0.25				0.3125				0.375				0.500			
	MIN	0.245				0.3065				0.369				0.493			
ACROSS F FLATS	MAX	0.500				0.562				0.625				0.812			
	MIN	0.489				0.551				0.612				0.798			
ACROSS G CORNERS	MAX	0.577				0.650				0.722				0.936			
	MIN	0.557				0.628				0.698				0.910			
HEAT H HEIGHT	MAX	0.135				0.198				0.260				0.327			
	MIN	0.125				0.188				0.250				0.312			
THREAD M LENGTH	REF	0.750				0.875				1.000				1.250			
UNTHREADED N BODY LGT /2	MAX	0.125	0.089			0.139	0.104			0.156	0.104			0.192	0.125		
L LENGTH	FINISH 1/	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX	DASH NO.		BODY LGT MIN	GRIP LGT MAX
3.750	CAD	1537	1565	1.477	2.250	1591	1619	1.200	2.000	1695	1721	0.625	1.500	1743	1767	0.071	1.000
	PHOS	1538	1566			1592	1620			1696	1722			1744	1768		
	ZNNI-K16	1050	1150			1063	1163			1074	1174			1085	1185		
	ZNNI-K20	2050	2150			2063	2163			2074	2174			2085	2185		
4.000	CAD	1539	1567	1.727	2.500	1593	1621	1.450	2.250	1697	1723	0.875	1.750	1745	1769	0.321	1.250
	PHOS	1540	1568			1594	1622			1698	1724			1746	1770		
	ZNNI-K16	1051	1151			1064	1164			1075	1175			1086	1186		
	ZNNI-K20	2051	2151			2064	2164			2075	2175			2086	2186		
4.500	CAD	1541	1569	2.227	3.000	1595	1623	1.950	2.750	1699	1725	1.375	2.250	1747	1771	0.821	1.750
	PHOS	1542	1570			1596	1624			1700	1726			1748	1772		
	ZNNI-K16	1052	1152			1065	1165			1076	1176			1087	1187		
	ZNNI-K20	2052	2152			2065	2165			2076	2176			2087	2187		
5.000	CAD									1701	1727	1.875	2.750	1749	1773	1.321	2.250
	PHOS									1702	1728			1750	1774		
	ZNNI-K16									1077	1177			1088	1188		
	ZNNI-K20									2077	2177			2088	2188		

/2 See Requirement 6.
/1 See Requirement 8.

TABLE II. Length tolerance.

NOMINAL SIZE	.250 .3125 .375	.4375 .500	.5625 .625 .750	.875 1.000	1.250
LENGTH	TOLERANCE (+.000)				
THRU 1.000	-.030	-.030	-.030		
OVER 1.000 THRU 2.500	-.040	-.060	-.080	-.100	-.120
OVER 2.500 THRU 4.000	-.060	-.080	-.100	-.140	-.160
OVER 4.000				-.160	-.180

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REQUIREMENTS:

1. Material: Steel, SAE Grade 8 in accordance with procurement specification.

2. Protective coating:

   Cadmium/1 plated in accordance with SAE AMS-QQ-P-416, Type II, Class 2.
   Zinc phosphate coated in accordance with MIL-DTL-16232, Type Z, Class 2.
   Zinc-Nickel plated in accordance with MIL-PRF-32647, Type K16.
   Zinc-Nickel plated in accordance with MIL-PRF-32647, Type K20.

/1 NOTE: Both cadmium and hexavalent chromium are R1, R2, and D1 substances in the Aerospace and Defense Declarable Substance List. Zinc-Nickel per MIL-PRF-32647, Type K16 is a preferred alternate for Cadmium plating and MIL-PRF-32647 Type K20 is preferred alternate for Zinc plating/Hexavalent Chromium passivate.

3. Threads: Threads shall be in accordance with FED-STD-H28.

4. Dimensions: All dimensions are in inches.

5. Head dimensions: For head dimensions see procurement specifications.

*6. Unthreaded length: Bolts too short for full thread length shall have threads extending over the full length. The distance from the underside of the head to the first complete (full form) thread, as measured with a ring thread gage, shall not exceed the maximum unthreaded - body length "N".

7. Head markings: Heads shall be marked on top with the manufacturer's identification and the number 8 to designate Grade 8 material. Raised marking are preferred.

8. Part number: The MS part number consists of the MS number, plus the dash number.

   Example: MS35764-1201 = 0.250 x 0.250 x 0.075 x 20 threads per inch, Cadmium plated, Type II, Class 2.
   Example: MS35764-1202 = 0.250 x 0.250 x 0.750 x 20 threads per inch, Zinc Phosphate coated, Type Z, Class 2.
   Example: MS35764-1001 = 0.250 x 0.250 x 0.750 x 20 threads per inch, Zinc-Nickel plated, Type K16.
   Example: MS35764-2001 = 0.250 x 0.250 x 0.750 x 20 threads per inch, Zinc-Nickel plated, Type K20.

9. Referenced documents shall be of the issue in effect as of the date of invitation for bid.

10. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

11. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TABLE III. Grade 8 – (*Reference information only*).

NOMINAL SIZE	THREADS PER INCH	PROOF LOAD LBS (MIN)	RECOMMENDED AXIAL LOAD (LBS)	RECOMMENDED TIGHTENING TORQUE */ (FT – LBS)
.250	20	3800	2850	11
	28	4350	3263	14
.3125	18	6300	4725	21.5
	24	6950	5213	27
.375	16	9300	6975	40.5
	24	10550	7913	55
.500	13	17050	12788	102
	20	19200	14363	126
.625	11	27100	20325	212
	18	30700	23025	263
.750	10	40100	30075	390
	16	44750	33563	464
1.000	8	72700	54525	889
	12	79550	59663	1050
1.250	7	116300	87225	1956
	12	128750	96563	2253

NOTES:

1. Torque based on 0.2 coefficient of friction.
2. Torque and axial load based upon 75% of proof load value.
3. The self-locking feature of these bolts is full effective when tightened to the torque values shown herein. Lockwashers or locknuts are not needed. (See note 5)

*/ 4. Torque values are to be shown adjacent the bolt part number callouts upon assembly drawings. They are subject to upward or downward revision by the agency responsible for the assembly drawings when the tightening fraction coefficient for specific applications is found to vary from as shown in (1) above.

5. Where material softer than the bolt material is fastened, plain washer of material as hard as the bolt of adequate thickness and larger in diameter than the bolt shall be used under the bolt head.

(TABLE V - Interchangeability Table. Obsolete and has been DELETED.)

Custodians:
Army - AT
Navy - OS
Air Force - 99
DLA - CQ

Preparing Activity:
AT
(Project 5306-2021-019)

Review Activities:
Army – CR4, MI
Navy – MC

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